Combinational logic circuit

In a simple copy machine, a stop signal, S, is to be generated to stop the machine operation and energize an indicator light whenever either of the following conditions exists: (1) there is no paper in the paper feeder tray; or (2) the two microswitches in the paper path are activated, indicating a jam in the paper path. The presence of paper in the feeder tray is indicated by a HIGH at logic signal P. Each of the microswitches produces a logic signal (Q and R) that goes HIGH whenever paper is passing over the switch to activate it. Design the logic circuit to produce a HIGH at output signal S for the stated conditions.

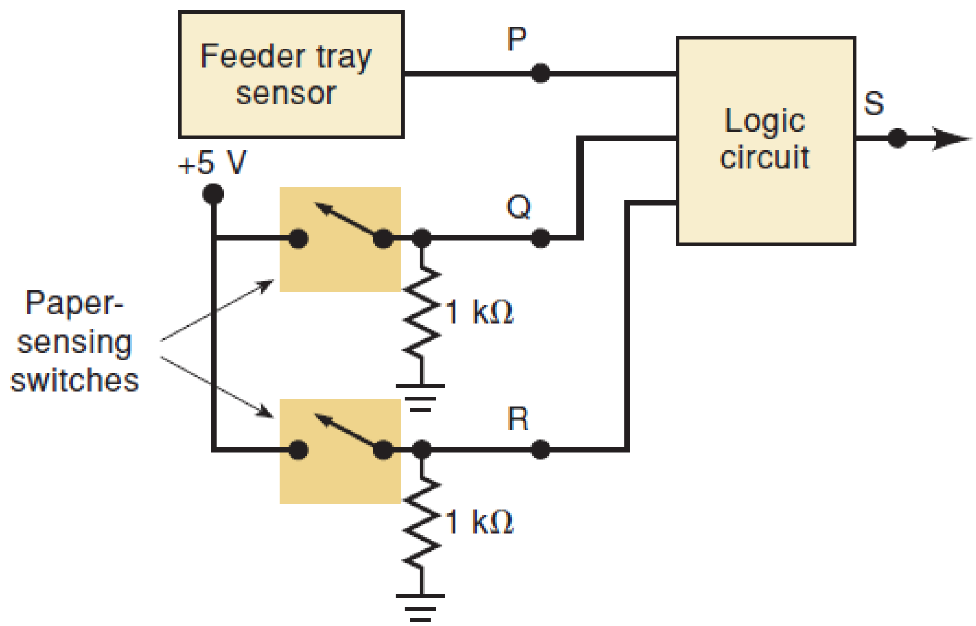


Figure 1

Use this information to write a Boolean expression for the truth table using AND, OR and NOT gates.

You then need to:

i. Construct a schematic diagram and simulate the operation of the logic circuit using standard logic gates.

ii. Minimise the logic circuit using a Karnaugh Map, and then construct a schematic diagram, and simulate the operation of the minimised logic circuit, using standard logic gates.

iii. Use De Morgan's Theorems to convert the minimised circuit to use only NAND gates.

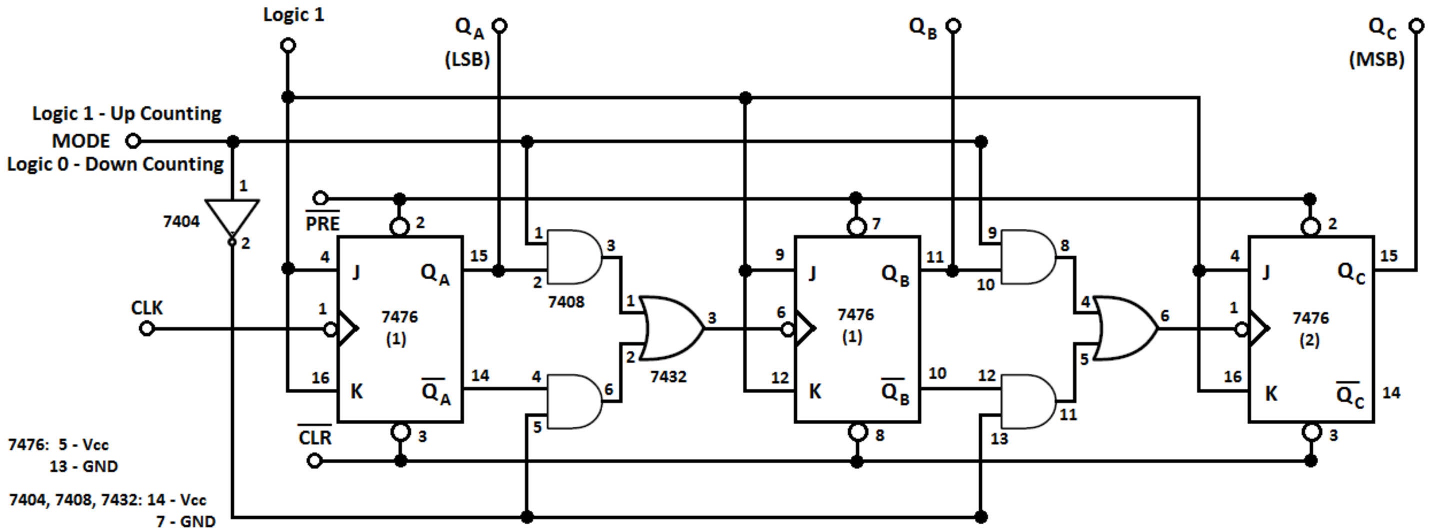
iv. Build a prototype of your final circuit and verify that it meets the conditions in the truth table, and then compare the results from theory, simulation and measurement.

Title: 3 bit asynchronous Up/Down counter using flip flops.

Components: IC 7476 (-Ve edge triggered Dual J-K Flip-Flops), IC 7408 (Quad 2-input AND gates), IC 7432 (Quad 2-input OR gates), IC 7404 (Hex Inverters), LEDs, Resister (220 Ω).

Equipment: Regulated DC power supply (0-25V), DMM, Breadboard, Connecting wires

Circuit Diagram:



Note: *Connect series combination of 220Ω resistor and LED between output and ground to see output.*

Observation Table:

Up Counter

Clock Input	Output		
Count	Q_C	Q_B	Q_A
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

Down Counter

Clock Input	Output		
Count	Q_C	Q_B	Q_A
7	1	1	1
6	1	1	0
5	1	0	1
4	1	0	0
3	0	1	1
2	0	1	0
1	0	0	1
0	0	0	0

Timing Diagram:

A.Up Counter: **B.Down Counter:**

You should:

•Identify the pin out of a IC7476 dual JK flip-flop with positive edge trigger (or specify an equivalent that you have available)

•Construct the schematic circuit of a 3 bit asynchronous Up/Down counter using flip flops using NI multisim. The IC7476 in the schematic has Set (PRE) and Reset (CLR) pins held low (logic 0) to allow changes to clock through to the output Q. (or give equivalent detail for JK available.)

(i)Use the schematic to simulate the behavior of the circuit by using CLK as a manual clock.

(ii)Build a prototype circuit and verify that it works as expected, and then compare the results from theory, simulation and measurements.

(iii)What practical improvements would you make to the circuit?

(iv)Draw a schematic diagram to show how logic gates can be added to the Up/Down counter in the diagram to convert it into an up counter.

(v)Simulate the circuit designed in to demonstrate how the circuit can be used as an up or down counter.

Precautions:

1. Always connect ground first and then connect Vcc.
2. The kit should be off before changing the connections.
3. Switch off the kit after the experiment.

Set up:

A. For Up Counting:

4. Connect the circuit as shown in the diagram.
5. Connect $\overline{PRE}$ input to the logic 1 i.e. +5V.
6. Connect $\overline{CLR}$ input to the logic 0 i.e. 0V or ground to reset counter.
7. Connect $\overline{CLR}$ input to the logic 1.
8. Connect Mode input to logic 1 for Up counting.
9. Apply the clock pulse to CLK input.
10. Observe the output and verify the observation table.

B. For Up Counting:

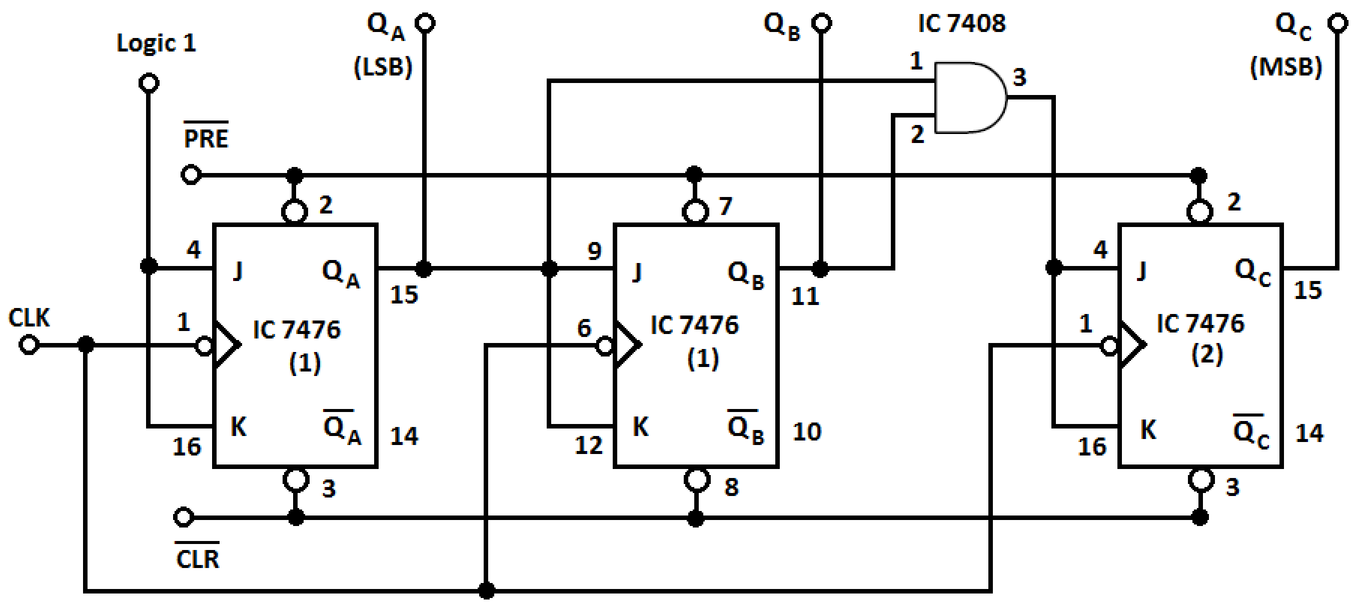
1. Connect the circuit as shown in the diagram.
2. Connect $\overline{CLR}$ input to the logic 1 i.e. +5V.
3. Connect $\overline{PRE}$ input to the logic 0 i.e. 0V or ground to set counter.
4. Connect $\overline{PRE}$ input to the logic 1.
5. Connect Mode input to logic 0 for Down counting.
6. Apply the clock pulse to CLK input.
7. Observe the output and verify the observation table.

Title: 3-bit Synchronous Counter using Flip-Flops.

Components: IC 74LS76A (-Ve edge triggered Dual J-K Flip-Flops), IC 7408 (Dual I/P Quad AND gates).

Equipments: Regulated DC power supply (0-25V), DMM, Breadboard, Connecting wires

Circuit Diagram:



7476: 5 - Vcc, 13 - GND

***Note:* 1.** *Connect series combination of 220Ω resistor and LED between output and ground to see output.*

2. *IC 7473 (Dual JK flip-flop with reset; negative-edge trigger) can also use instead of IC 74LS76A.*

Observation Table:

Clock Input	Output		
Count	Q_C (2^2)	Q_B (2^1)	Q_A (2^0)
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

Timing Diagram:

You should:

• Identify the pin out of a IC7476 dual JK flip-flop with negative edge trigger (or specify an equivalent that you have available)
• Construct the schematic circuit of a 3-bit Synchronous Counter using Flip-Flops using NI multisim. The IC7476 in the schematic has Set (PRE) and Reset (CLR) pins held low (logic 0) to allow changes to clock through to the output Q.

i.Use the schematic to simulate the behavior of the circuit by setting data values at the input and clocking them through.
ii.Build a prototype of the circuit and demonstrate that it works as expected, and then compare the results from theory, simulation and measurements
iii.What practical improvements would you make to the circuit?
iv.Draw a schematic diagram to show how logic gates can be added to the Up counter in the diagram to convert it into an up/down counter.
v.Simulate the circuit designed in to demonstrate how the circuit can be used as an up or down counter.

Precautions:

1. Always connect ground first and then connect Vcc.
2. The kit should be off before changing the connections.
3. Switch off the kit after the experiment.

Procedure:

1. Connect the circuit as shown in the diagram.
2. Connect $\overline{PRE}$ input to the logic 1 i.e. +5V.
3. Connect $\overline{CLR}$ input to the logic 0 i.e. 0V or ground to reset counter.
4. Connect $\overline{CLR}$ input to the logic 1.
5. Apply the clock pulse to CLK input.
6. Observe the output and verify the observation table.